Investigative action report INV-4098-U1

HIGH SEVERITY

Investigate U1 Iddq failure on BRD-264098

Prepared for Lumina Silicon from manufacturing, test and supply-chain data in Splunk by Cisco.

Organization
Lumina Silicon

Report
INV-4098-U1

Severity
High

Work order
WO-INV-5001

Priority
P1

Assignee
Equipment engineer

Due
Within 2 days

Board
BRD-264098 (REVIEW)

Chassis
SYS-CH-886A-B

Component
U1 Application SoC (silicon die)

Part number
SOC-7N-A07

Supplier and lot
Northgate Semi, 2631-552A

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SUMMARY

1 of 3 measurements on U1 are out of limit on Lumina Silicon board BRD-264098. The data points to reflow thermal excursion stressed the part (78 % confidence).

PROBLEM STATEMENT

Iddq measured 342.0 µA against a limit of at most 300.0 µA.

KEY INDICATORS

Primary measurement
Iddq 342.0 µA, limit at most 300.0 µA

Margin to limit
-14 %, out of limit

Population percentile
100th, of 180 recent boards

Strongest driver
r = 0.87, Chamber peak temperature

Retests
3 attempts, 3 failed

Chamber peak
227.1 °C, limit 224.5 °C

Minimum N2 flow
482 sccm, limit 455 sccm

Component lot exposure
13 boards, 4 in review, lot 2631-552A

Same tool in process
2, boards on CVD-01

Supplier quality
83 DPPM, 98.2 % on time, 5 in review

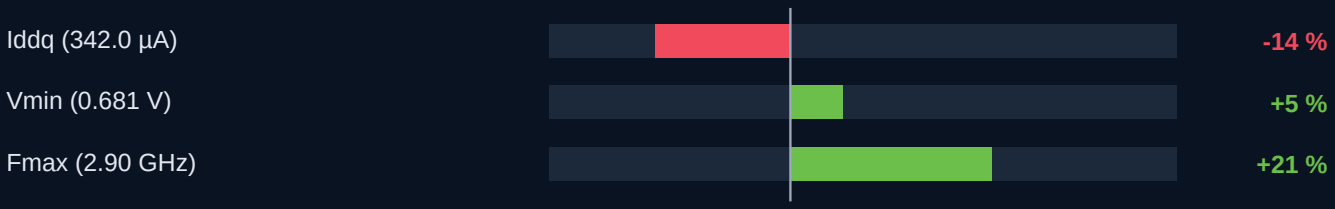
EVIDENCE

- Iddq is 342.0 µA against a limit of at most 300.0 µA (14 % over limit, 100th percentile of 180 boards).
- Vmin is 0.681 V against a limit of at most 0.720 V (5 % margin, 51th percentile of 180 boards).
- Fmax is 2.90 GHz against a limit of at least 2.40 GHz (21 % margin, 51th percentile of 180 boards).
- The board saw a chamber peak of 227.1 °C (limit 224.5) and a minimum N2 flow of 482 sccm (limit 455).
- Strongest process correlation: Chamber peak temperature (r = 0.87).
- The failure repeated on 3 attempts, so a fixture or contact fault is unlikely.

- Northgate Semi has 83 DPPM, 98.2 % on-time delivery and 5 boards in review.
- The last four weekly lots of Iddq drift toward the upper limit, so the shift started before this board.
- 13 boards were built from component lot 2631-552A and 4 are in review; 1 more board is in process on CVD-01.
- The process window spent 7 s above the 224.5 °C limit.

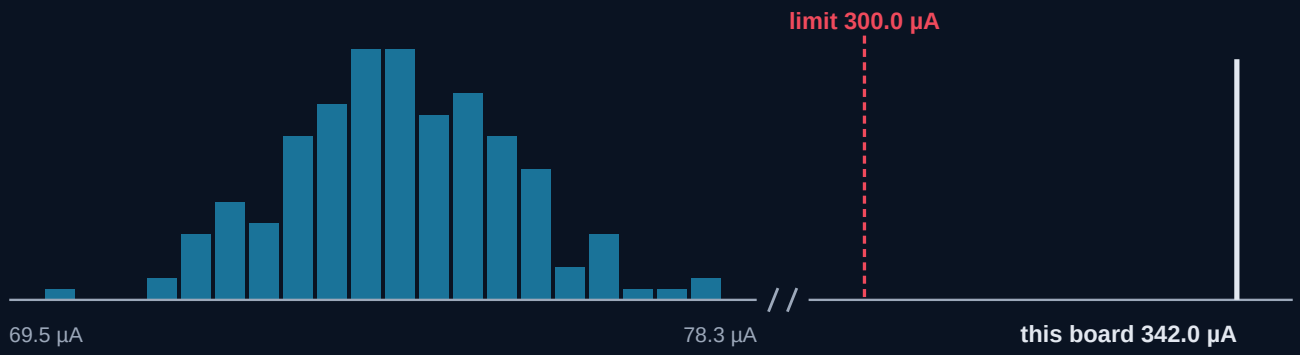
Evidence charts

Margin to limit by measurement



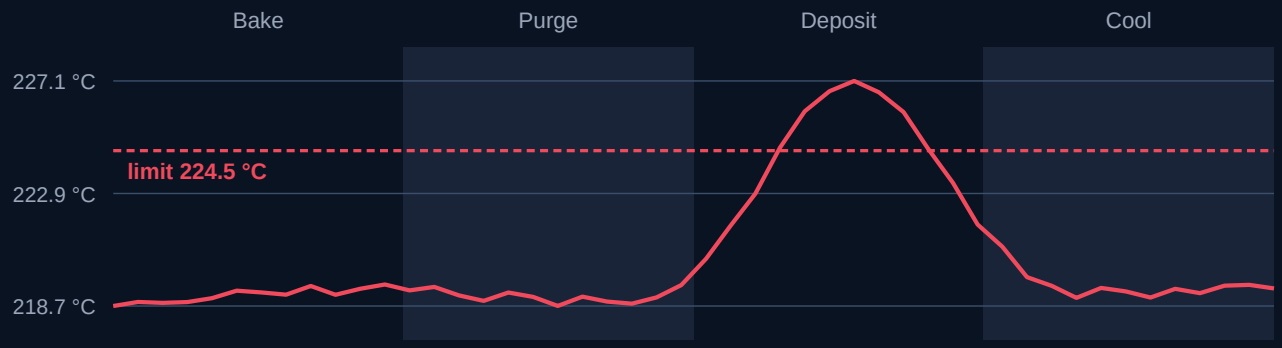
How far each reading sits inside its limit. Negative values are out of limit.

Iddq across 180 recent boards



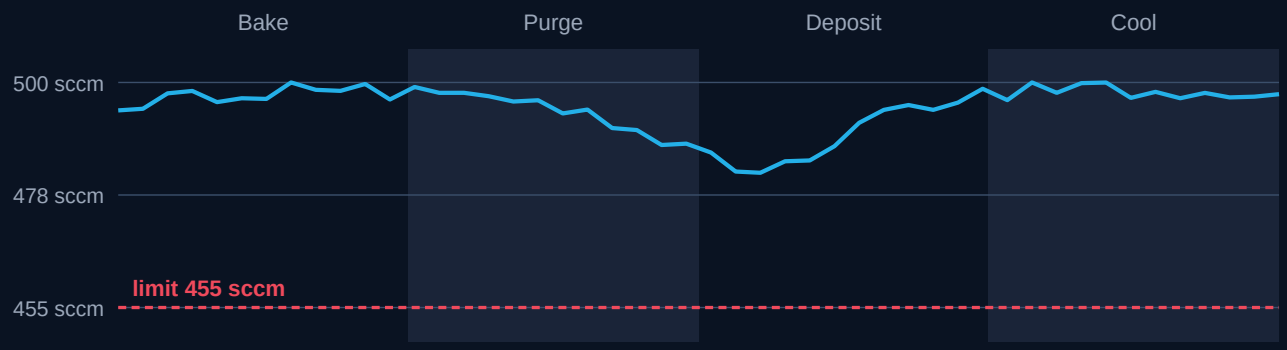
This board is at the 100th percentile and far outside the population, more than 6 standard deviations from the mean. Population mean 74.1 µA, standard deviation 1.32 µA. The axis is broken so the population spread stays readable.

Chamber temperature during this board's process window



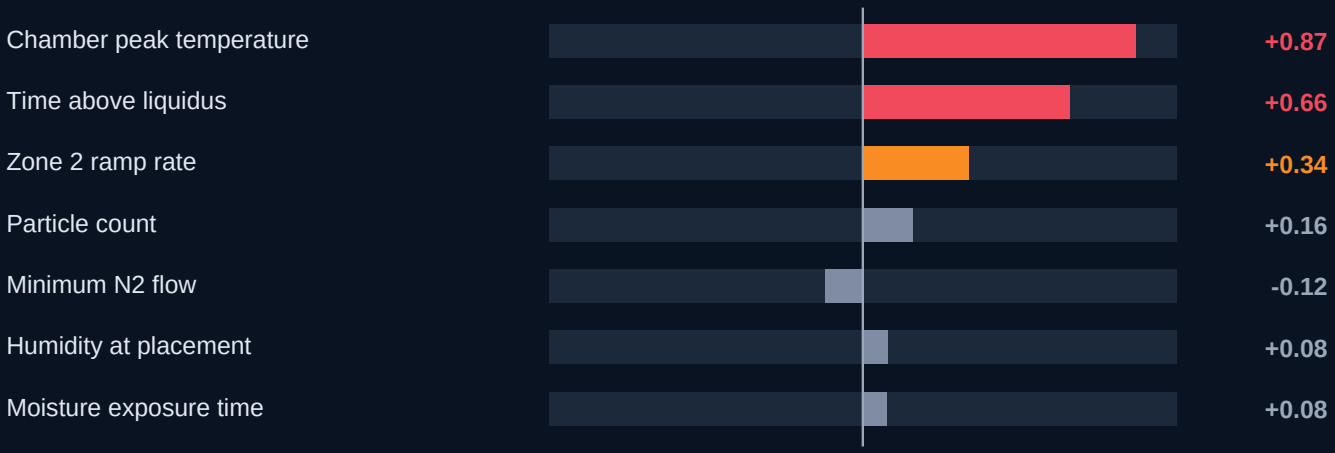
Peak 227.1 °C against the 224.5 °C limit, 7 s above the limit. Reconstructed from the logged peak and setpoint because per-board samples are not retained.

N2 purge flow during this board’s process window



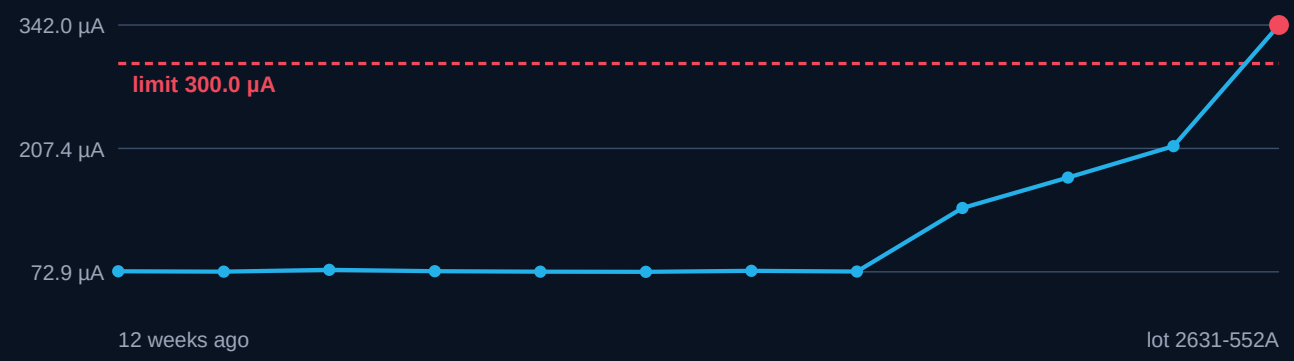
Minimum 482 sccm against the 455 sccm limit, 0 s below the limit. Reconstructed from the logged minimum and setpoint.

Correlation of Iddq with process variables



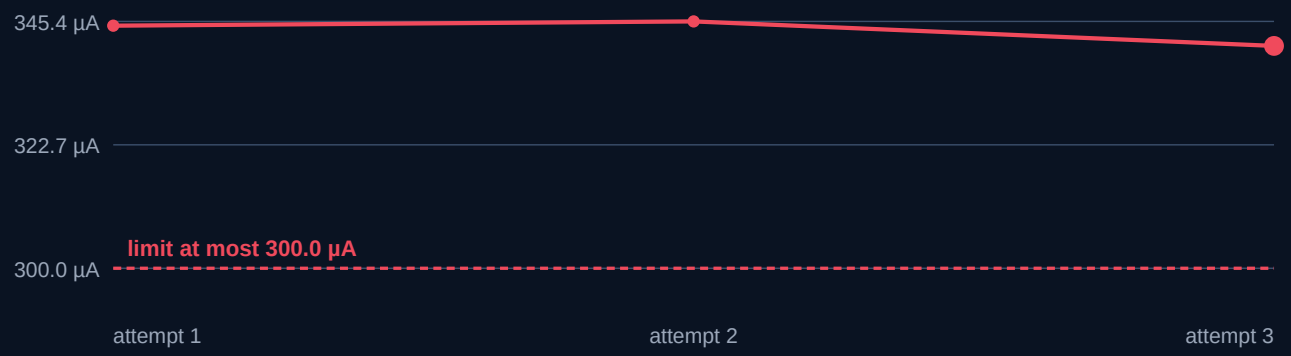
Pearson r across the board population. Above 0.6 in either direction is a strong association, which is not proof of cause.

Iddq by weekly lot, last 12 weeks



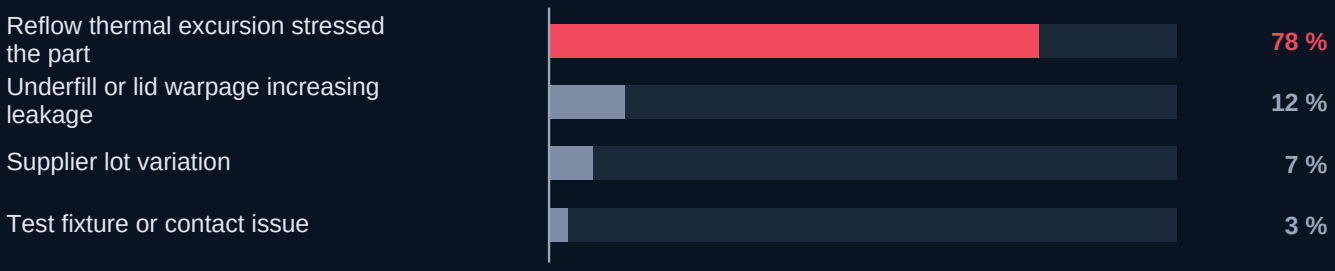
The last four lots drift toward the upper limit and this board’s lot 2631-552A is out of limit.

Retest repeatability, Iddq



3 attempts on FCT-03 spread 1.3 %. Consistent readings point to a real defect rather than fixture or contact noise.

Root-cause confidence



Ranked by the analyst from the measurements, correlations, retests and supplier history.

Root cause and actions

PROBABLE ROOT CAUSES

Cause	Confidence	Rationale
Reflow thermal excursion stressed the part	78 %	Chamber peak 227.1 °C is over the limit and correlates at r = 0.87 with the failing reading.
Underfill or lid warpage increasing leakage	12 %	Warpage margin is thin on this package family at high peak temperatures.
Supplier lot variation	7 %	Recent lots show a drift toward the limit.
Test fixture or contact issue	3 %	Retests repeat within a few percent and the golden unit passed.

CONTAINMENT

- Hold board BRD-264098 and chassis SYS-CH-886A-B from shipment.
- Quarantine remaining stock of lot 2631-552A and the boards built in the same work order window.
- Check boards still in process on CVD-01 for the same excursion.

CORRECTIVE ACTIONS

Action	Owner	Due
Inspect zone heater and thermocouple TC-ZN2 on CVD-01 and recalibrate	Equipment engineering	Within 2 days
X-ray solder joints and run thermal cycling on this part	Failure analysis	Within 3 days
Request an 8D report from Northgate Semi if lot variation is confirmed	Supplier quality	Within 7 days

VERIFICATION

- Retest the board after the corrective action and confirm all measurements are in limit.
- Confirm the next two boards on CVD-01 show no excursion.
- Close the 8D when three consecutive lots are stable.

Work order WO-INV-5001

Investigate U1 Iddq failure on BRD-264098

Priority P1, assigned to Equipment engineer, due within 2 days. Estimated effort 12 hours.

#	Task	Detail	Hours
1	Contain affected boards and stock	Hold board BRD-264098 and chassis SYS-CH-886A-B from shipment. Quarantine remaining stock of lot 2631-552A and the boards built in the same work order window. Check boards still in process on CVD-01 for the same excursion.	1
2	Verify tool condition	Inspect zone heater and thermocouple TC-ZN2 on CVD-01 and recalibrate	3
3	Physical failure analysis	X-ray solder joints and run thermal cycling on this part	6
4	Disposition and report	Rework, scrap or release the board and update the 8D record.	2
	Total		**12**

APPROVALS

Role	Name	Signature	Date
Engineering			
Quality			
Production			